

Advanced Structural Dynamics and Aeroelasticity

Lecture 1 Introduction to *Dynamics of Rotor Systems*

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Personal intro and contact

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Unit and theme organisation

- Acronyms:
 - ASDA ... Advanced Structural Dynamics and Aeroelasticity
 - DRS ... Dynamics of Rotor Systems
- ASDA unit (20cp)
 - TB2 ASDA exam (70%)
- DRS theme:
 - ~16-18 lectures
 - DRS theme labs and report (15%)
- Resources and feedback:
 - Blackboard (slides, example sheets, ...)
 - TB2 1st half feedback via open office hours
 - TB2 2nd half feedback via lab-based coursework

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Primary applications: helicopters

Photographs taken by unknown authors from Internet.



AgustWestland AW101



Boeing A160 Hummingbird



Airbus X3



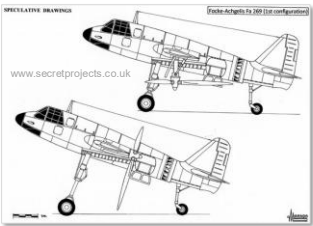
Sikorsky X2

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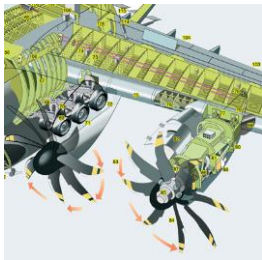
Primary applications: tiltrotors



Primary applications: eVTOL



Secondary applications: fix wing propellers



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Secondary applications: jet propulsion

Roll Royce Ultrafan
www.rolls-royce.com/innovation/ultrafan.aspx



Safran Open Rotor
mentourpilot.com



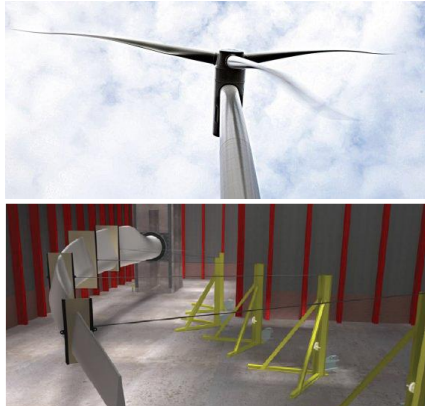
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Tertiary applications: wind turbines

Photographs taken by unknown authors from Internet.



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Learning objectives

- To develop an understanding of aerospace structures with rotors with particular focus on the **dynamic properties** of rigid and elastic rotors and their aeroelastic interaction with flexible support structures.
- DRS theme learning objectives:
 - develop **mathematical models** of aerospace structures with rotors and propellers;
 - analyse **dynamic properties** of rotating flexible blades and elastically suspended rigid propellers;
 - evaluate **aeroelastic response and stability** characteristics of coupled propeller-wing systems.
- DRS topics:
 - Topic 1: Rotating blade
 - Isolated rotating flapping or lead-lagging rigid and flexible blades *in vacuo*
 - Topic 2: Elastically suspended propeller
 - Elastically suspended rigid propeller disc with *quasi-steady aero disturbances*

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Useful resources and general horizon expansion ...

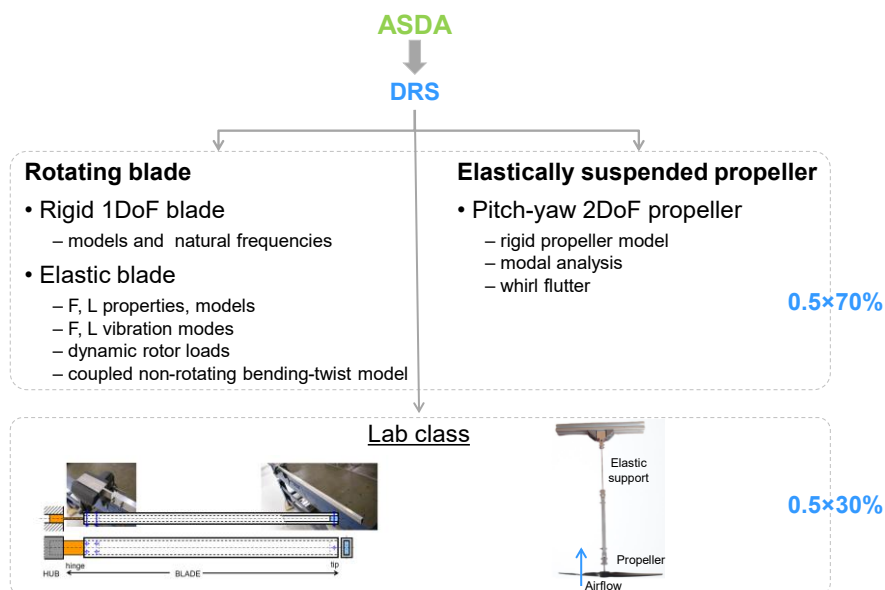
- Books
 - Bielawa, Richard L. *Rotary wing structural dynamics and aeroelasticity*. American Institute of Aeronautics and Astronautics, 2006.
 - Bramwell, Anthony Robert Southey, David Balmford, and George Done. *Bramwell's helicopter dynamics*. Elsevier, 2001.
 - Čečrdle, Jiří. *Whirl flutter of turboprop aircraft structures*. Elsevier, 2023.
 - Johnson, Wayne. *Helicopter theory*. Courier Corporation, 2012.
- Papers and reports
 - Selection of useful and interesting papers will be provided via Blackboard and released gradually on weekly basis.
- Remarks
 - These books are not essential to make expected progress in DRS. However, using or referencing some of them, *according to your own needs and style*, can help you to tackle tricky parts and expand your general knowledge.

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Overview of Dynamics of Rotating Systems



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Technical challenges in *Dynamics of Rotating Systems*

- **Dynamics** issues severely restrict operation of **flexible components**.
- Combination of symmetric design and asymmetric operations causes **periodic loading** and **resonant responses**.
- Unfavourable **dynamic coupling** can cause **loss of stability**.
- Important considerations:
 - Centrifugal and gyroscopic effects
 - Dependencies on the speed of rotation
 - Dynamic and aeroelastic interactions
 - Mechanical complexity to enable control
- Design drivers:
 - Functionality (conceptual design, control and handling, architecture)
 - Statics (loads, mass, balance, etc.)
 - **Dynamics** (comfort, resonance, gusts, landing, etc.)
 - **Stability** (ground and air stability, flutter, divergence, etc.)
 - Fatigue (closing the loop back to Dynamics)
 - Noise (closing the loop back to Functionality)

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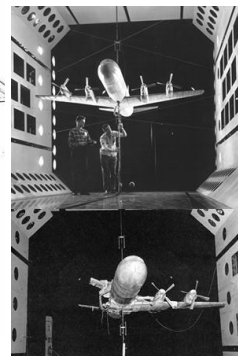
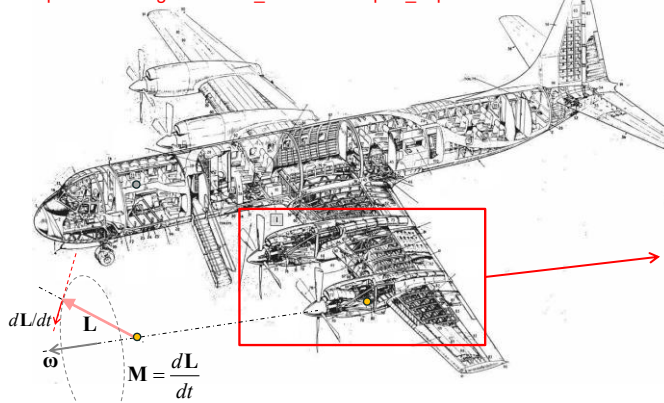
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Technical challenges in *elastically suspended props*

Lockheed L-188 Electra

- Braniff Airways Flight 542, N9705C, Buffalo, Texas; September 29, 1959
- Northwest Airlines Flight 710, N121US, Cannelton, Indiana; March 17, 1960

https://www.faa.gov/lessons_learned/transport_airplane/accidents/N9705C



What happens if "... the stiffness of the supporting system is reduced through failed or damaged engine structure, mounts ... or nacelle structure ..."?

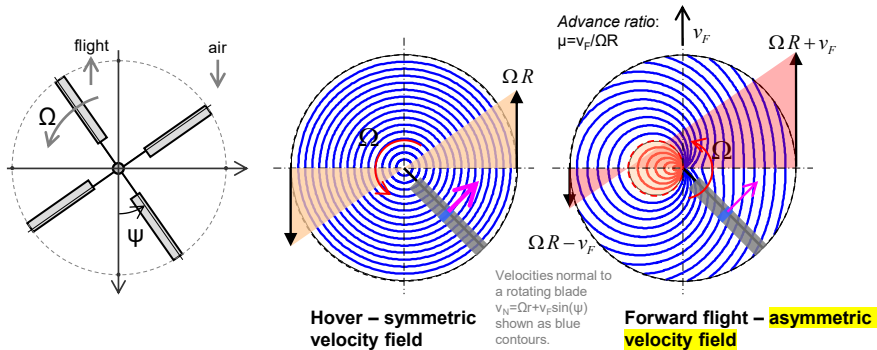
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Technical challenges in rotating blades

While *helicopter rotors* are symmetric systems, the velocity and flow fields that they produce in forward (or any steady directional) flight are asymmetric. Quasi-symmetry of the flow is present only during axial (vertical) flight or hover.



$$\text{Lift} \sim v^2, \quad v = v(r, \psi)$$

Flow asymmetry is responsible for **periodically repeating blade loading** (i.e. dynamic loading with the period $T = 2\pi/\Omega$). This condition causes large oscillatory bending moments in the blade root and large rolling moment. Flow asymmetry is present even after the trimmed conditions are established, e.g. forward flight. Powerful periodic blade loading is also responsible for the resulting blade motion.

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Technical challenges in rotating blades

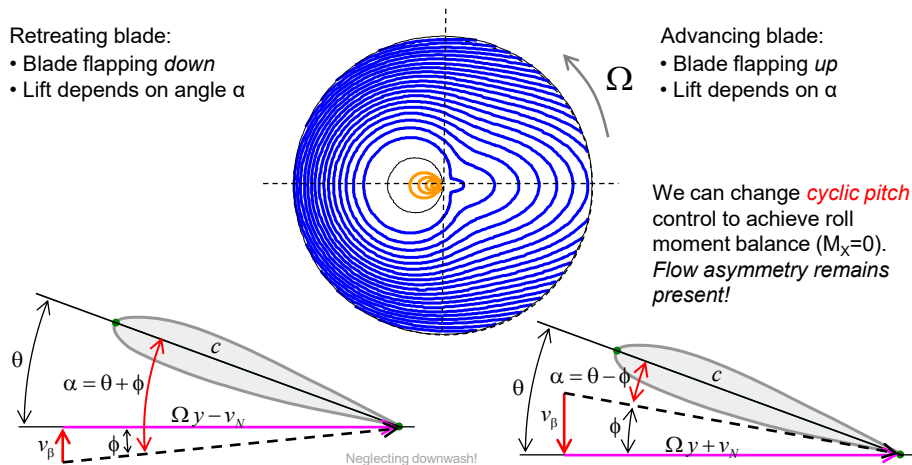
We can observe **blade flapping** (flapping angle β) due to flow asymmetry which then causes periodically changing flow incidence or *effective Angle of Attack* (AoA) α .

Retreating blade:

- Blade flapping *down*
- Lift depends on angle α

Advancing blade:

- Blade flapping *up*
- Lift depends on α



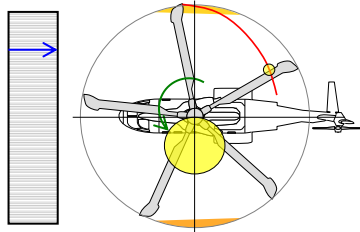
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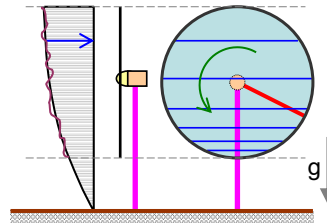
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Other examples of “operational asymmetry”

Multiple sources of loads can be identified in rotors during their steady operation. We distinguish: (a) loads which are fundamentally linked with operational principles (e.g. flow asymmetry), (b) loads linked with operational issues (e.g. manufacturing problems, failure, wear, setting, etc.).



- flow asymmetry during directional flight
- asymmetry due to coning (longitudinal)
- Blade-Vortex Interaction (BVI)
- wake-structure, tail rotor, ...
- Unbalance
 - mass distribution
 - aerodynamics (airfoil, pitch, ...)



- vertical wind profile (shear)
- Tower “shadow”
- Gravity
- gusts, waves, turbulence ...
- Unbalance
 - mass distribution
 - aerodynamics (airfoil, pitch, ...)

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Classic response to rotating blade challenges ...

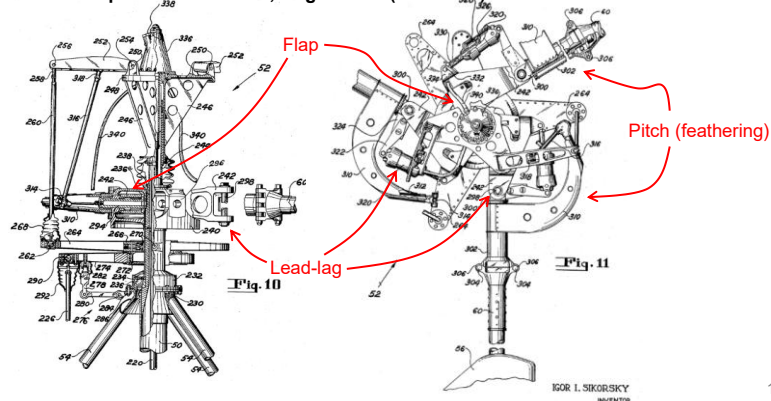
To address the challenges with forced flapping, resulting blade lead-lag and the need to control induced aero loads via blade pitch adjustments, Igor Sikorsky introduced:



November 1939
(1st configuration by Igor Sikorsky)

Credits: www.sikorskyarchives.com

Patent US 2517509 Helicopter Rotor Control, Aug. 1 1950 (filed 1945)



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Helicopter rotor hub complexity

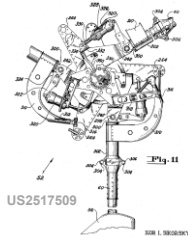
Comparison between early, classical and contemporary hubs:



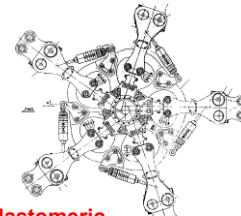
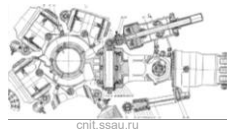
www.sikorskyarchives.com



www.turbosquid.com



mechanically articulated hub



elastomeric articulated hub

Rotor hub systems:

- **articulated** (VS-300), **hingeless** (Lynx), **bearingless** (EC135), **elastomeric** (AW101).
- teetering (R44), gimbal (AW609), stiff (S-97), rigid (Karem's OSTR).
- Propeller and wind turbine rotor hub systems – pitch control with rigid flap-lag.

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Blade pitch control

Blade pitch control enables stable and controlled helicopter flight. Pitch angle θ for each blade is fully controlled and can be described as a function $\theta=f(\psi)$.

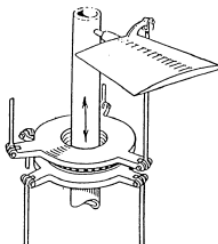
This approach controls changes in *Thrust* generated by the rotor including its orientation and offset between its line of action and the helicopter centre of mass.

Technical implementation of this concept is based on the use of tilting (*cyclic pitch*) and translating (*collective pitch*) swashplates and pitch links.

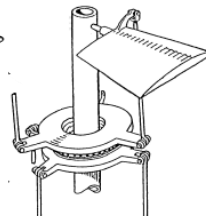
$$\theta(\psi) = \theta_0 + a_1 \sin(\psi) + b_1 \cos(\psi) + a_2 \sin(2\psi) + b_2 \cos(2\psi) + \dots$$

Standard pitch control

"Higher" harmonic pitch control



Collective (translation)



Cyclic (tilt)



Active Pitch Link (NASA, 2010)

Other pitch control methods:

- Westland Lynx spider mechanism
- Kaman (K-MAX, SH-2G) servo-flap
- swashplateless, active twist, ...

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Summary

- Recognise challenges affecting *dynamics of rotating systems*
 - Gyroscopic effects,
 - Periodic and resonant forcing, etc.
- Understand the emergence of “asymmetry” across mentioned *rotor system contexts* (flow, operation, design).
- Appreciate the origins and design implications associated with rotor operating conditions (complexity, mass, cost).
- Study the diversity of rotor hub design solutions:
 - Flexibility of the rotor support system and its resilience to failure.
 - Requirements to accommodate blade flapping, lead-lag, and pitch.
 - Introduce dynamic protections measures (e.g., dampers).
 - Rotor operation and performance control (pitch, speed, etc.).